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PHYSICS-INFORMED NEURAL NETWORKS FOR REMAINING-USEFUL-LIFE PREDICTION OF ROLLING-ELEMENT BEARINGS

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Abstract

It is frequently the state of the condition of rolling element bearings which is a critical issue in the safe running of rotating machinery. The progression of a bearing fault may begin with a localized surface defect, followed by vibration growth, heat, misalignment, secondary damage and unplanned shutdown. Prediction of the remaining useful life is thus an important problem in prognostics and health management. In this thesis, bearing RUL prediction is investigated using a selected set of run-to-failure data from the NASA IMS bearing data set and the performance is compared to three data-driven baselines. This pipeline loads the IMS data from the local file, extracts time domain and envelope spectral features, creates normalized RUL targets, builds a PCA-based health indicator, and tests four different models: feed-forward neural baseline model, proposed DeepXDE physics-informed model, LSTM sequence model, and 1D CNN sequence model. The final evaluation is performed with 12 full-run train-validation-test sets and three different random seeds are used for each training set. Across 36 evaluated runs per model, the data-only feed-forward baseline achieved the lowest mean RMSE at 0.209 ± 0.156, followed by the proposed PINN at 0.216 ± 0.170, the LSTM at 0.230 ± 0.078, and the CNN at 0.235 ± 0.095. The standard deviations and block-bootstrap uncertainty intervals demonstrate marked changes in the results depending on the split configuration and held-out bearing run. The work, in other words, does not assert that it is statistically superior to any one of the models. Rather, it provides a repeatable local processing approach, a well documented feature and evaluation process, and a conservative analysis that suggests that the weak physics-informed formulation investigated is competitive in certain splits and not robustly superior in light of limited IMS run-to-failure data.

Keywords: remaining useful life, rolling-element bearing, prognostics and health management, physics-informed neural network, IMS bearing dataset, vibration analysis, LSTM, CNN.

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I also thank my family for their constant support during the long process of learning the theory, writing code, running experiments, correcting mistakes, and preparing this thesis. Their patience made the work easier to continue when the results did not come out as simply as expected.

Finally, I acknowledge the researchers who made the IMS bearing dataset publicly available. Open run-to-failure datasets make it possible for students and researchers to test prognostic methods in a reproducible way.

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Nomenclature

Symbol or term	**Meaning**
ANN	Artificial neural network
BPFO	Ball pass frequency outer race
BPFI	Ball pass frequency inner race
BSF	Ball spin frequency
CNN	Convolutional neural network
DeepXDE	Python library used for physics-informed neural networks
FFT	Fast Fourier transform
FNN	Feed-forward neural network
FTF	Fundamental train frequency
HI	Health indicator
IMS	Intelligent Maintenance Systems
LSTM	Long short-term memory network
MAE	Mean absolute error
PCA	Principal component analysis
PHM	Prognostics and health management
PINN	Physics-informed neural network
REB	Rolling-element bearing
RMS	Root mean square
RMSE	Root mean squared error
RUL	Remaining useful life
R^2	Coefficient of determination

Chapter 1

Introduction

1.1 BACKGROUND AND MOTIVATION

Rolling-element bearings support shafts, carry loads, and reduce friction in motors, pumps, turbines, gearboxes, fans, compressors, and many other machines. Because they sit directly in the load path, a bearing defect rarely remains an isolated issue for long. A small defect on a raceway or rolling element can create repeated impacts, excite vibration, produce heat, damage nearby parts, and eventually force the machine out of service. The consequences of a bearing failure are sometimes far greater than the cost of the bearing in industrial maintenance.

While traditional maintenance approaches could help, they don't completely address the issue. Reactive maintenance is waiting for failure and takes the risk of unplanned shutdown. Preventive maintenance involves exchanges of parts at a predetermined interval, typically a more conservative interval than necessary. Can take out healthy components too soon or miss faults that occur quicker than they should. The improvements of both methods are provided in condition-based maintenance, in that the state of the machine is assessed and a maintenance decision is made based on the evidence. Prognostics and health management is a further step that attempts to predict the remaining useful life of a component before failure, in addition to predicting failure, [1, 2].

The reason why the remaining useful life prediction for bearings is difficult is that the deterioration process is not linear. Vibration may stay low for a considerable length of the test, but then increase dramatically towards the end of life. The measured signals are affected by the type of fault, the speed of the fault, the load, the position of the sensor, noise, resonance, lubrication, and the geometry of the bearings [3, 4]. Training on one of the bearings doesn't mean that the model works well on another bearing, even if they are both from the same test rig. This is the reason that the IMS bearing data set is helpful for benchmarking run-to-failure experiments. They enable a study of the same raw measurements and the comparison of methods under identical conditions [5].

The deep learning techniques have gained popularity for their ability to capture nonlinear patterns within vibration signals, which is crucial for bearing diagnosis and RUL prediction. CNNs can learn local patterns from windows of data, LSTMs can model temporal dependence, and feed-forward networks can fit engineered feature vectors [6–8]. These methods can work well when the training and test data are similar. Their weakness is that they usually learn only from examples. When labeled run-to-failure data are limited or the test bearing degrades differently from the training bearings, a data-only model may produce predictions that are numerically plausible but physically questionable.

Physics-informed neural networks offer a way to include prior engineering knowledge in the learning process. In their most established form, PINNs train a neural network while penalizing violations of governing equations or physical constraints [9]. Bearing RUL prediction does not usually provide a complete closed-form degradation equation, but it does provide useful partial knowledge. Damage should generally increase with operating time, RUL should generally decrease, and bearing geometry gives characteristic fault frequencies associated with cage, inner-race, outer-race, and rolling-element defects. This thesis studies whether such weak physical information can improve RUL prediction on the IMS data.

1.2 PROBLEM STATEMENT

The problem studied in this thesis is the prediction of normalized remaining useful life for selected bearings in the IMS run-to-failure dataset. The main question is whether a physics-informed neural network using time and fault-frequency information can generalize better across bearing runs than comparable data-driven neural baselines.

This question is practical as well as methodological. In a real machine, the future failure path is unknown. A maintenance engineer may have historical data from one bearing, but the next bearing may not degrade in the same way. A useful RUL model must therefore do more than memorize one run. It must learn a relationship between measured condition and remaining life that transfers, at least partly, across bearings.

1.3 OBJECTIVES OF THE STUDY

The objectives of this thesis are:

1. To extract degradation-relevant vibration features from selected IMS bearing runs and construct normalized RUL targets with a PCA-based health indicator for analysis.
2. To design and evaluate a DeepXDE physics-informed neural model that uses weak monotonicity and fault-frequency energy priors for bearing RUL prediction.
3. To compare the proposed physics-informed model with data-only, LSTM, and CNN neural baselines under complete held-out bearing splits and interpret where each model succeeds or fails.

1.4 SCOPE

This study is limited to the IMS bearing data placed locally under `data/raw/1st_test`, `data/raw/2nd_test`, and `data/raw/3rd_test`. Four bearing runs are used in the final experiment design: `ds2_b1`, `ds1_b3`, `ds1_b4`, and `ds3_b3`. The work uses vibration signals only. It does not add temperature, oil debris, acoustic emission, or motor-current data.

The models predict normalized RUL from extracted features. The thesis does not claim field deployment readiness. The data come from a controlled test rig, and the operational conditions are much simpler than many industrial machines. The physics-informed part is intentionally weak: it relies on monotonicity and fault-frequency energy priors rather than a full bearing degradation law or crack growth model.

1.5 IMPORTANCE OF THE STUDY

This work connects three parts of the bearing prognostics problem in one local workflow: signal processing, machine learning, and physical interpretation. The feature extraction step keeps links with vibration analysis through RMS, kurtosis, crest factor, and envelope spectral-energy features. The neural models test whether those features can predict normalized RUL across complete bearing runs. The physics-informed model tests whether simple monotonicity and fault-frequency priors improve learning when run-to-failure data are limited.

The final result is intentionally conservative. The 12-configuration repeated-seed evaluation shows that the data-only feed-forward baseline has the lowest mean RMSE, while the proposed weakly physics-informed model remains close but less consistent across splits. This is still useful because it identifies where the current physical prior is insufficient and why more run-level validation is needed before making strong claims.

1.6 LIMITATIONS

The main limitations are:

1. The IMS dataset has a small number of total run-to-failure records, which creates an experiment design constraint.
2. The selected physics prior is not a full bearing degradation law. It is a regularizer based on monotonic RUL behavior and fault-frequency energy.
3. The model uses hand-engineered features, not raw vibration waveform.
4. The RUL labels are derived from elapsed time to the end of each run, which assumes the last available file represents failure or near-failure.

5. The experiments are run on one local PC, so training time and numerical results may vary on other machines.

1.7 THESIS OUTLINE

Chapter 2 reviews bearing failure, vibration-based condition monitoring, RUL prediction, deep learning baselines, and physics-informed learning. Chapter 3 describes the IMS dataset, feature extraction, preprocessing, health indicator construction, model architectures, and evaluation metrics. Chapter 4 describes the local implementation, experiment splits, software environment, and reproducibility setup. Chapter 5 presents the results and discusses model behavior across the 12 split configurations and repeated random seeds. Chapter 6 summarizes the findings, contributions, limitations, and future work.

Chapter 2

Literature Review

2.1 INTRODUCTION

This chapter reviews the research background for bearing RUL prediction with physics-informed neural networks. The discussion follows the structure of the thesis proposal, but it is updated to reflect the completed experiments. It begins with rolling-element bearing failure modes and vibration-based health monitoring, then reviews predictive maintenance methods, deep learning for bearing prognosis, general PINN theory, and recent physics-informed learning work in bearing condition monitoring. The chapter closes by identifying the research gap addressed in this thesis.

2.2 LITERATURE SURVEY

2.2.1 Rolling-Element Bearings: Fundamentals and Failure Modes

A rolling-element bearing contains an inner race, an outer race, rolling elements, and a cage. The rolling elements transfer load through small contact regions, so repeated stress cycles can initiate subsurface or surface-initiated fatigue. Defects may also arise from lubrication failure, contamination, overload, misalignment, electrical pitting, poor installation, or manufacturing variation. Once a defect begins, repeated rolling contact can grow it into a spall or crack that changes the vibration response of the machine [10–12].

The vibration signature depends on the component that is damaged. An outer-race defect produces impacts when rolling elements pass over a fixed damaged region. An inner-race defect rotates with the shaft and is affected by the load zone. Rolling-element and cage defects produce different modulation patterns. These mechanisms explain why bearing monitoring often uses characteristic frequencies derived from bearing geometry and shaft speed. The frequencies are not ideal fault labels but they define meaningful regions in the spectrum, and envelope spectrum [3, 13].

The thesis proposal highlighted that the vibration features are not arbitrary numbers, but a legitimate representation of the data and hence should be used as such in bearing prognostics. This is still relevant in the finished product. Even though these are subsequently used in a neural model, RMS, kurtosis, crest factor, and envelope energy all have a mechanical meaning. Physically interpretable features also enable a deeper understanding of why a model is successful or unsuccessful in a held-out run of bearings.

2.2.2 Traditional Predictive Maintenance Techniques for Bearings

Normally, predictive maintenance is part of condition-based maintenance and prognostics and health management. Jardine et al. [1] discuss the overall machinery diagnostics and prognostics, and Lei et al. [2] review the entire process of data acquisition to RUL prediction. The typical process for a bearing application covers sensor measurement, preprocessing, feature extraction, construction of health indicators, diagnosis of the faults and estimation of future life.

Vibration remains one of the most widely used signals for bearing health monitoring because bearing faults generate impacts and modulations in the mechanical response. Time-domain features such as RMS and kurtosis summarize energy and impulsiveness. Frequency-domain and envelope-domain features can reveal periodic impact patterns linked to bearing kinematics. Similarity-based and health-indicator methods then convert these features into a degradation measure or RUL estimate [14].

The IMS bearing dataset is one of the common public run-to-failure datasets used for this type of work. It is associated with the wavelet-filter bearing prognostics study of Qiu et al. [5]. The PRONOSTIA platform is another well-known accelerated bearing degradation benchmark [15]. Public datasets are valuable because they let different researchers test methods on the same raw measurements. At the same time, public datasets are small compared with industrial variability, so the train-test split must be designed carefully. Random snapshot splits can overstate performance by allowing information from the same run-to-failure trajectory into both training and testing.

2.2.3 Deep Learning in Predictive Maintenance

Deep learning has become popular in bearing prognostics because it can fit nonlinear relationships between sensor measurements and degradation labels. A feed-forward neural network can work directly with engineered feature vectors. A CNN can detect local patterns from a sequence or transformed signal representation. An LSTM can model temporal dependence and is therefore naturally suited to degradation trajectories. Babu et al. [6] showed early CNN-based RUL regression, and later studies extended deep learning for transfer learning, high-quality representation learning, and hybrid sequence architectures [7, 8, 16].

Recent studies also combine convolutional, recurrent, and autoencoder structures for bearing health monitoring and RUL prediction [17–19]. These models are attractive because inference is fast after training and because they can learn patterns that are difficult to specify manually. However, their performance depends strongly on the similarity between the training and test distributions. In bearing RUL prediction, this is a serious issue because two bearings under similar nominal conditions may fail at different times and with different feature trajectories.

For this reason, model evaluation should hold out complete bearing runs when the research question is transfer across bearings. A random split of individual snapshots may measure interpolation within one degradation curve rather than generalization to another bearing. The experiment design in this thesis uses complete held-out runs for validation and testing.

2.2.4 Physics-Informed Neural Networks

Physics-informed neural networks were introduced as a way to train neural networks under physical residual constraints [9]. In the usual PINN setting, a neural network approximates a field variable, and automatic differentiation is used to compute derivatives that appear in a governing equation. The training loss combines data mismatch and a physics residual. This has made PINNs attractive in fluid mechanics, heat transfer, wave propagation, and inverse problems where governing equations are available.

The body of PINN literature has been growing rapidly. Lawal et al. [20] and Ren et al. [21] provide reviews of methodological advances and optimization problems, as well as applications in scientific computing. The power of a PINN isn't just a neural network and a physical equation. It is dependent on the relevance of the physical residual, its correct weighting, and its numerical trainability. Inappropriate physical limitations can hinder training or bias the results.

Bearing RUL prediction is not a textbook PINN problem because the degradation process is not governed by a single known partial differential equation. Rolling contact fatigue, lubrication, load distribution, resonance, material variation, and measurement noise interact in ways that are difficult to reduce to one residual. Crack-growth laws such as Paris and Erdogan [22] show how mechanics can describe damage propagation in some contexts, but a complete bearing RUL equation for vibration features is rarely available in practice. This motivates weaker physics-informed learning: monotonic damage behavior, lifetime distributions, fault-frequency consistency, and other engineering priors.

2.2.5 Physics-Informed Neural Networks for Bearing Predictive Maintenance

Recent bearing-related studies have begun to use physics-informed or knowledge-informed learning. Chen et al. [23] proposed a physics-informed deep neural network for bearing prognosis with multisensory signals. Parziale et al. [24] applied PINNs to condition monitoring of rotating shafts. Herwig et al. [25] developed a signal-processing and physics-informed network for explainable bearing condition monitoring. Zhong et al. [26] used a multi-source physics-informed SincNet structure for rolling bearing fault diagnosis. von Hahn and Mechefske [27] investigated a knowledge-informed loss based on Weibull lifetime behavior.

These studies show why the approach is promising. Physics can improve interpretability, guide learning when data are limited, and discourage predictions that conflict with engineering expectations. They also show why the approach must be tested carefully. A physical prior only helps when it matches the measured data closely enough. If the prior is too weak, the model may behave like a data-only network. If it is too strong or mismatched, it may prevent the model from fitting valid degradation patterns.

2.3 RESEARCH GAP

The literature shows three gaps that motivate this thesis. First, many bearing RUL studies report strong results, however their preprocessing, RUL labeling, train-test split, and evaluation protocol differ. This makes comparisons difficult. Second, some studies use random sample splits that can leak run-specific degradation information into both training and testing. Third, physics-informed learning for bearing prognosis is still developing, and there is no guarantee that a weak physical prior improves prediction on complete held-out bearing runs.

This thesis addresses a narrower and reproducible question: how does a weak physics-informed RUL model behave when compared with common neural baselines on complete held-out IMS bearing runs? The study uses a local project structure, explicit feature extraction, a defined RUL target, saved tables and figures, 12 train-validation-test split configurations, and repeated random seeds. The goal is not to claim a universal best model. The goal is to test a specific physics-informed formulation and report what the completed experiment shows.

Chapter 3

Dataset and Research Methodology

3.1 INTRODUCTION

This chapter describes the dataset, signal-processing steps, RUL target construction, model definitions, experiment design, and evaluation metrics used in the study. The workflow begins with raw IMS vibration files and ends with model rankings, prediction curves, and error analysis. Figure 3.1 summarizes the local pipeline.

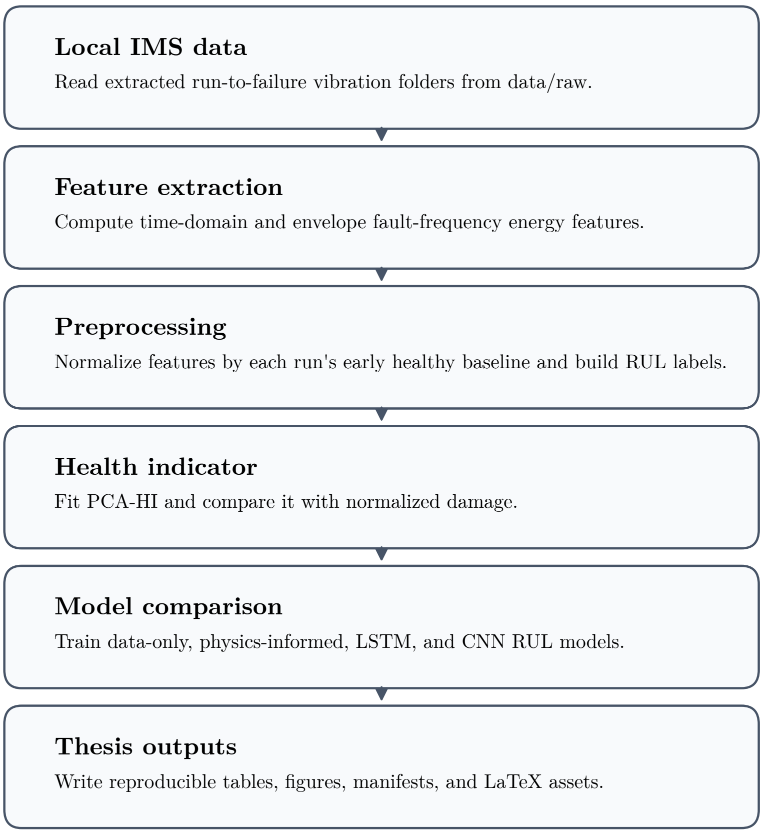

Figure 3.1: End-to-end local workflow used for IMS bearing RUL prediction.

3.2 EXPERIMENTAL SETUP

The IMS bearing data were generated by the NSF I/UCR Center for Intelligent Maintenance Systems with support from Rexnord Corporation. The test rig used four Rexnord ZA-2115 double-row bearings mounted on a common shaft. An AC motor kept the shaft speed constant at 2000 RPM, and a spring loading mechanism applied a radial load of 6000 lb to the shaft and bearings. The bearings were force lubricated throughout the run-to-failure tests. The IMS README also notes that the observed failures occurred after the bearings exceeded their designed lifetime of more than 100 million revolutions.

Vibration was measured at the bearing housings using PCB 353B33 High Sensitivity Quartz ICP accelerometers. The first data set used two accelerometers per bearing, placed in the x- and y-axis directions. The second and third data sets used one accelerometer per bearing. Data collection was performed with an NI DAQ Card 6062E. Figure 3.2 shows the bearing test rig and sensor placement used for the IMS experiments.

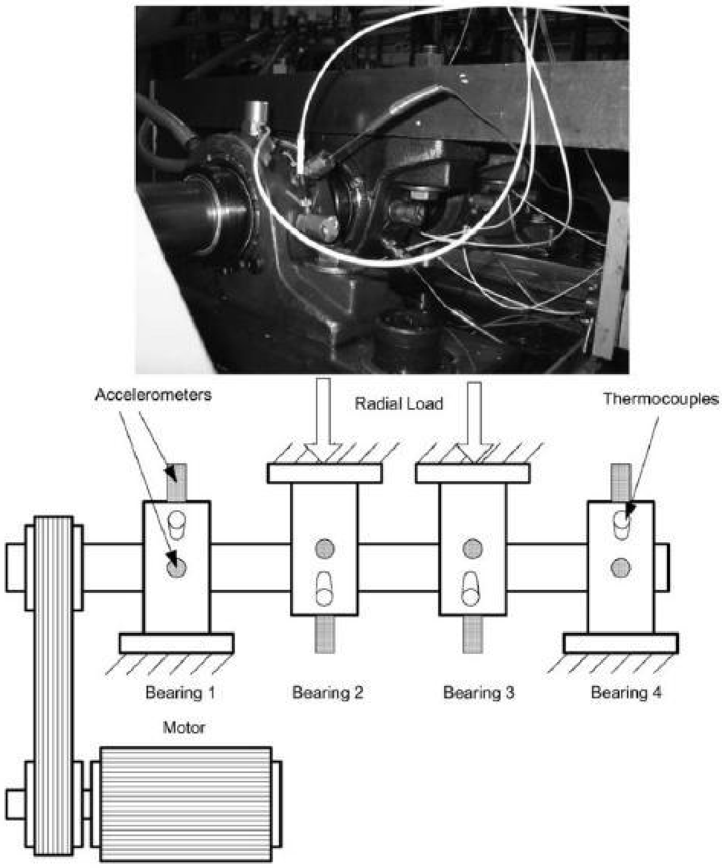

Figure 3.2: IMS bearing test rig and sensor placement used for run-to-failure data collection, adapted from the IMS Bearing Data README and Qiu et al. [5].

3.3 DATASET DESCRIPTION

The IMS bearing dataset contains run-to-failure vibration measurements collected from the setup described above. The raw data are stored as timestamped ASCII text files. Each file contains a nominal one-second vibration snapshot with 20,480 data points sampled at 20 kHz. The filename records the collection time, and larger gaps between timestamps indicate that the experiment resumed on the next working day. In this project, the extracted dataset folders are stored locally under `data/raw`.

The local pipeline uses three IMS folders: `1st_test`, `2nd_test`, and `3rd_test`. The first test contains eight columns because two accelerometer channels are available for each of four bearings. The second and third tests contain four columns, one for each bearing. The implementation maps each selected column to a bearing and averages two axes when the first test provides x and y directions for the target bearing.

Table 3.1: Bearing runs used in the final experiments.

Run ID	IMS dataset	Bearing number	Number of samples	Used in experiments
ds2_b1	2nd_test	1	984	Used as test, validation, and training run across the 12 split configurations
ds1_b3	1st_test	3	2156	Used as test, validation, and training run across the 12 split configurations
ds1_b4	1st_test	4	2156	Used as test, validation, and training run across the 12 split configurations
ds3_b3	3rd_test	3	6324	Used as test, validation, and training run across the 12 split configurations

Figure 3.3 shows the number of snapshots available for each selected run. The difference in run length is large. The third dataset bearing run has 6324 samples, while the second dataset bearing run has 984 samples. This imbalance affects model training. The final 12-configuration split design therefore uses balanced sampling within the training runs so that the longest run does not dominate the training set.

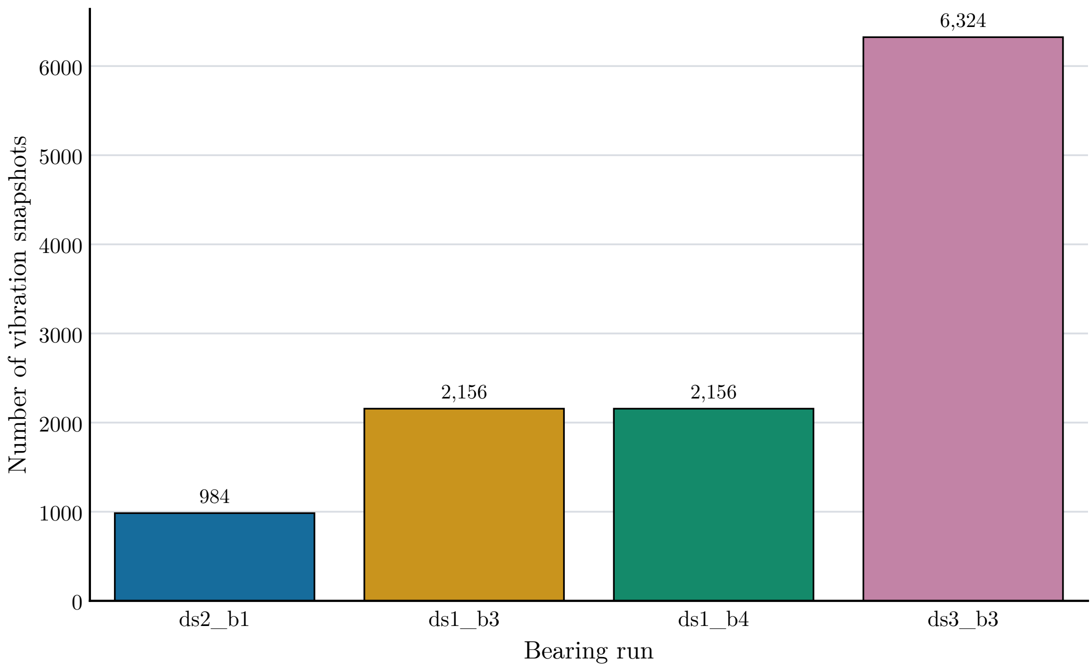

Figure 3.3: Snapshot coverage for the four bearing runs used in the final experiments.

3.4 FEATURE EXTRACTION

Each vibration snapshot is converted into a feature vector. The time-domain features are RMS, standard deviation, peak-to-peak value, kurtosis, crest factor, and mean absolute value. These features summarize amplitude, spread, impulsiveness, and peak behavior. They are widely used in bearing condition monitoring because they are simple and physically interpretable.

The physics-related features are calculated from envelope spectral energy near bearing fault frequencies. The signal is centered, transformed with the Hilbert transform to obtain the envelope, centered again, windowed with a Hann window, and transformed with a real FFT. For each fault frequency, the pipeline sums spectral energy in a 5 Hz band around the first four harmonics. The four energy features are E_{FTF}, E_{BPFO}, E_{BPFI}, and E_{BSF}. Their sum is stored as E_{kin}.

The fault-frequency equations used by the implementation are

$$FTF = \frac{f_r}{2}\left(1 - \frac{d}{D}\cos\theta\right), \tag{3.1}$$

$$BPFO = \frac{nf_r}{2}\left(1 - \frac{d}{D}\cos\theta\right), \tag{3.2}$$

$$BPFI = \frac{nf_r}{2}\left(1 + \frac{d}{D}\cos\theta\right), \tag{3.3}$$

$$BSF = \frac{Df_r}{2d}\left(1 - \left(\frac{d}{D}\cos\theta\right)^2\right), \tag{3.4}$$

where f_r is shaft frequency, n is the number of rolling elements, d is rolling-element diameter, D is pitch diameter, and θ is contact angle.

Table 3.2: Extracted features used by the neural models.

Feature name	Domain	Physical meaning	Used by models
elapsed_scaled	time feature	Normalized elapsed operating time used as temporal input	yes
rms	time-domain	Signal energy amplitude through root mean square vibration	yes
std	time-domain	Vibration dispersion around the mean	yes
ptp	time-domain	Peak-to-peak vibration range	yes
kurtosis	time-domain	Impulsiveness of vibration signal	yes
crest_factor	time-domain	Peak amplitude relative to RMS	yes
mean_abs	time-domain	Mean absolute vibration amplitude	yes
E_FTF	physics-informed frequency feature	Envelope spectral energy around cage fault train frequency harmonics	yes
E_BPFO	physics-informed frequency feature	Envelope spectral energy around outer-race fault frequency harmonics	yes
E_BPFI	physics-informed frequency feature	Envelope spectral energy around inner-race fault frequency harmonics	yes
E_BSF	physics-informed frequency feature	Envelope spectral energy around rolling-element fault frequency harmonics	yes
E_kin	physics-informed frequency feature	Combined bearing kinematic fault-frequency spectral energy	yes

Table 3.3: Bearing fault frequencies used for envelope spectral-energy extraction.

Frequency name	Symbol	Value in Hz	Related bearing component	Extracted feature
FTF	FTF	14.78	Cage / fundamental train	E_FTF
BPFO	BPFO	236.40	Outer race	E_BPFO
BPFI	BPFI	296.93	Inner race	E_BPFI
BSF	BSF	139.92	Rolling element	E_BSF

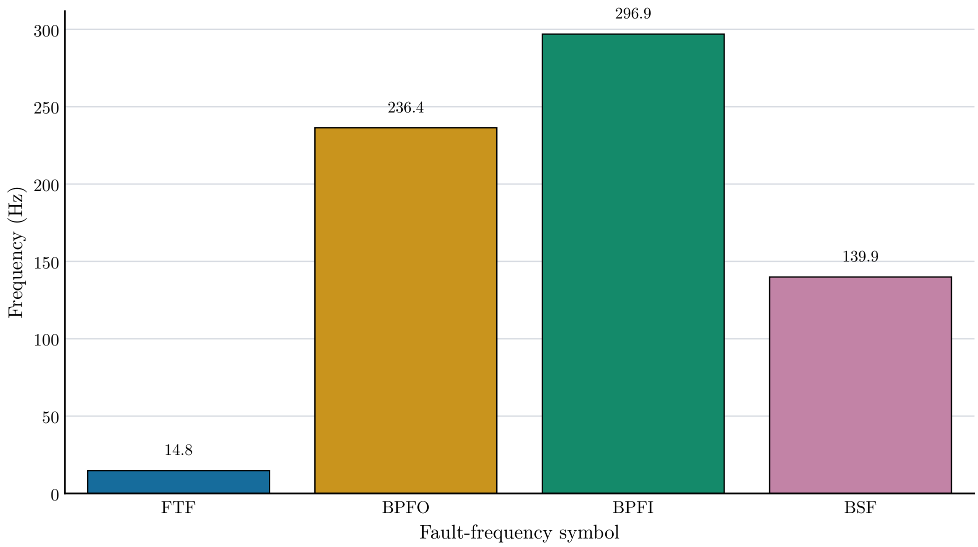

Figure 3.4: Bearing fault-frequency values used as markers for envelope spectral-energy extraction.

3.5 RUL TARGET LABELING

For each selected bearing run, elapsed time is computed from the timestamp of the first snapshot. RUL is computed by subtracting the current timestamp from the timestamp of the final snapshot. The model target is normalized RUL:

$$RUL_{norm}(t) = \frac{RUL(t)}{\max(RUL)}. \tag{3.5}$$

This gives a target near 1 at the beginning of the run and near 0 at the end. The normalization makes results comparable across runs with different durations. It also means that the model predicts relative life fraction, not absolute hours.

3.6 PREPROCESSING

Feature values can differ greatly between bearing runs. To reduce the effect of run-specific scale, each run is normalized using its early healthy baseline. The first 5 percent of samples, with a minimum of 20 samples, are treated as the healthy baseline. For each model feature, the median baseline value is used. The feature is converted to a nonnegative relative increase, transformed with `log1p`, and smoothed with a rolling median of 15 samples.

Elapsed time is converted to `elapsed_scaled` by dividing elapsed hours by 1200. The model input vector contains `elapsed_scaled` and the normalized feature set, including the fault-frequency energy features.

3.7 PCA HEALTH INDICATOR

A PCA-based health indicator is built to inspect whether the extracted features contain a monotonic degradation signal. The model feature matrix is standardized and projected onto the first principal component. The resulting component is scaled between 0 and 1. If the indicator direction is inverted, it is flipped so that higher values correspond to greater damage.

The health indicator is not used as the final RUL target. It is used as an explanatory tool. It helps show whether the engineered features broadly track degradation.

Table 3.4: PCA health-indicator monotonicity.

Bearing run	Monotonicity score	Interpretation
ds1_b3	0.766	Strongly degradation-consistent PCA-HI behavior
ds1_b4	0.691	Moderately or weakly degradation-consistent PCA-HI behavior
ds2_b1	0.803	Strongly degradation-consistent PCA-HI behavior
ds3_b3	0.864	Strongly degradation-consistent PCA-HI behavior

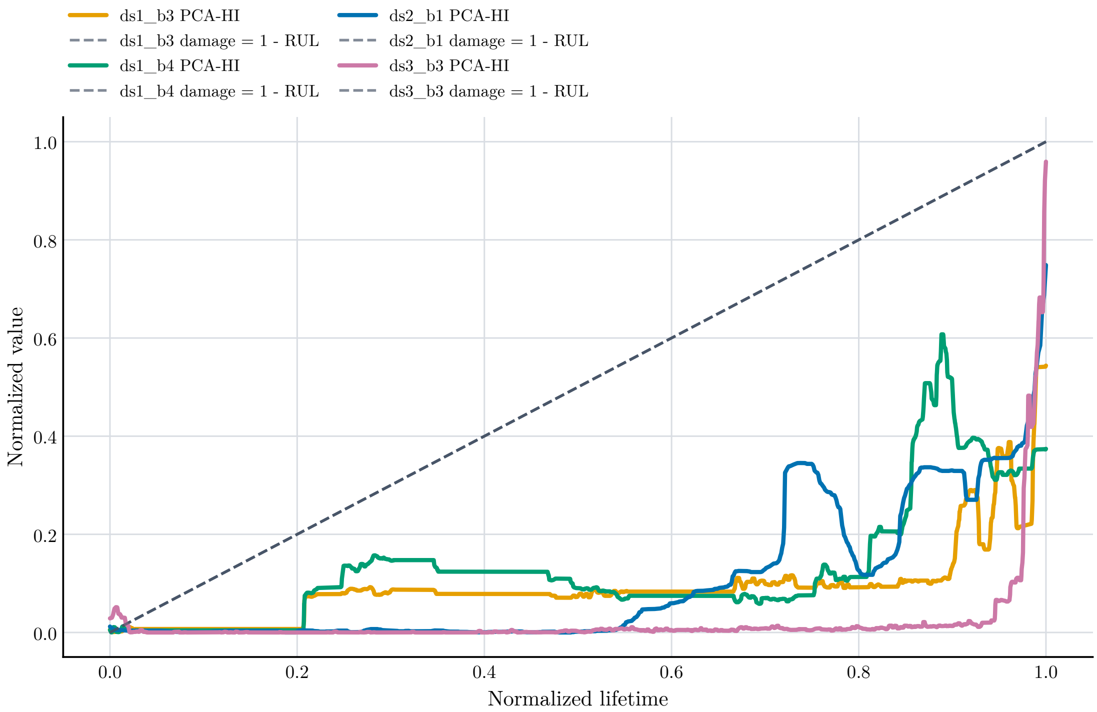

Figure 3.5: Smoothed PCA health indicator compared with normalized damage on held-out test runs.

3.8 MODEL DEFINITIONS

Four models are evaluated. The data-only neural baseline is a feed-forward neural network implemented in DeepXDE [28]. It uses three hidden layers with 64 neurons per layer and tanh activation. A sigmoid output transform constrains predictions between 0 and 1. It is trained with mean squared error.

The proposed physics-informed model uses the same basic feed-forward architecture but adds weak physical residuals. The first residual penalizes positive RUL slope with respect to time beyond a small tolerance, because RUL should generally decrease as life is consumed. The second residual connects high E_{kin} to damage by encouraging the predicted damage term $1 - RUL$ to be consistent with fault-frequency energy. The model loss can be written as

$$\mathcal{L} = \mathcal{L}_{data} + \lambda_m \mathcal{L}_{mono} + \lambda_e \mathcal{L}_{energy}, \tag{3.6}$$

where $\mathcal{L}_{data}$ is the supervised RUL loss, $\mathcal{L}_{mono}$ is the monotonicity residual, $\mathcal{L}_{energy}$ is the fault-energy residual, and λ_m and λ_e are small regularization weights.

The LSTM baseline uses sliding windows of feature vectors. It reads a sequence of length 20 and predicts the RUL at the end of the sequence. The architecture has one LSTM layer with 64 hidden units followed by a small dense head and a sigmoid output. The CNN baseline uses the same sequence windows but applies 1D convolutions across the time axis.

Table 3.5: Model definitions and roles in the experiment.

Model	Model type	Input format	Role in thesis	Target
Data-only neural baseline	Feed-forward neural network	Single snapshot feature vector	Data-only baseline	Normalized RUL
Proposed DeepXDE Physics-Informed RUL Model	DeepXDE physics-informed neural network	Single snapshot feature vector with time and fault-frequency energy features	Proposed model	Normalized RUL
LSTM baseline	Recurrent neural network	Sliding-window sequence of feature vectors	Sequence baseline	Normalized RUL
CNN baseline	1D convolutional neural network	Sliding-window sequence of feature vectors	Sequence baseline	Normalized RUL

3.9 EXPERIMENT DESIGN

The final evaluation uses 12 complete-run train-validation-test configurations. In each configuration, one bearing run is held out for testing, one different run is used for validation, and the remaining two runs are used for training. This covers all choices of test run and validation run among the four selected bearing trajectories. The design is stricter than randomly splitting snapshots because the model must transfer across complete bearing degradation trajectories. Each model is trained three times per split using different random seeds. Therefore, each model has 36 evaluated runs in the final statistical summary.

Table 3.6 and Figure 3.6 show the split design.

Table 3.6: Twelve complete-run train-validation-test split configurations.

Split	Train runs	Validation run	Test run
S01	ds1_b4 + ds3_b3	ds1_b3	ds2_b1
S02	ds1_b3 + ds3_b3	ds1_b4	ds2_b1
S03	ds1_b3 + ds1_b4	ds3_b3	ds2_b1
S04	ds1_b4 + ds3_b3	ds2_b1	ds1_b3
S05	ds2_b1 + ds3_b3	ds1_b4	ds1_b3
S06	ds2_b1 + ds1_b4	ds3_b3	ds1_b3
S07	ds1_b3 + ds3_b3	ds2_b1	ds1_b4
S08	ds2_b1 + ds3_b3	ds1_b3	ds1_b4
S09	ds2_b1 + ds1_b3	ds3_b3	ds1_b4
S10	ds1_b3 + ds1_b4	ds2_b1	ds3_b3
S11	ds2_b1 + ds1_b4	ds1_b3	ds3_b3
S12	ds2_b1 + ds1_b3	ds1_b4	ds3_b3

Figure 3.6: Experiment split matrix showing how each bearing run is used in the 12 configurations.

3.10 EVALUATION METRICS

Evaluation of the models is carried out using MAE, RMSE and R^2. MAE is the mean of absolute prediction errors. R^2 allows for larger errors to have a greater impact, since the error is squared before being averaged. The negative R^2 indicates that the model is worse at predicting the target than predicting the mean value of the target.

The thesis employs RMSE as the primary ranking criterion because large RUL errors are undesirable for maintenance planning. Still, MAE and R^2 are reported to provide a complete picture of model behavior. The metric definitions follow common practice in prognostics evaluation [29]. Statistical validation is reported as the mean and standard deviation across 12 split configurations and three repeated seeds per split. Prediction uncertainty is also estimated with 95% block-bootstrap intervals over contiguous prediction-error blocks within each held-out test trajectory.

Chapter 4

Local Implementation and Experimental Setup

4.1 LOCAL PROJECT STRUCTURE

This was the original work originally created in Google Colab notebook. The workflow was adapted to a local Python project for this thesis. Google Drive is no longer mounted in the code. It fetches data from `data/raw`, stores the extracted features in `data/processed_features` and outputs tables and figures to `outputs`.

The main package is under `src/thesis_work`. It includes individual modules for configuration, IMS data loading, feature extraction, preprocessing, model training, the metrics, report generation, and running the program from the command line. This format allows for the experiment to be easily tested and rerun, as opposed to a single notebook.

4.2 REPRODUCIBILITY COMMANDS

The project uses `uv` for dependency management. The main commands are:

```
uv sync --extra dev
uv run thesis-work validate-data
uv run thesis-work extract-features
uv run thesis-work run
uv run thesis-work regenerate-figures
uv run pytest -q
```

The full run trains all four models for 12 split configurations and three random seeds per split. The command can take a long time on CPU, so the project also supports shorter smoke-test arguments. The final repeated-seed rows are saved in `outputs/tables/split_seed_results_table.csv`, and the per-split mean results are saved in `outputs/tables/final_results_table.csv`.

4.3 SOFTWARE ENVIRONMENT

The implementation uses Python 3.11 and the following main libraries: NumPy, pandas, SciPy, scikit-learn, matplotlib, seaborn, DeepXDE, PyTorch, and tqdm. The LaTeX thesis is compiled with MiKTeX using the Windows batch script included in the thesis source folder.

The local PC used for the completed run had an AMD Ryzen 7 5800H CPU with 8 cores and 16 logical processors, 16 GB RAM, Windows 11 Home Single Language, and an NVIDIA GeForce RTX 3060 Laptop GPU with 4 GB VRAM. Training and inference wall time is machine-specific and should be reported together with this hardware context.

4.4 DATA VALIDATION

The validation command checks that the expected folders are present and that the first snapshot in each dataset can be read with the correct number of columns. This step matters because the IMS text files may be stored either as extracted folders or zip files. The local loader supports both, but the final run used extracted folders under `data/raw`.

4.5 FEATURE CACHE

Feature extraction is the slowest preprocessing step because every vibration snapshot must be read and transformed. The pipeline therefore caches per-run feature tables and a combined `all_runs_features.csv`. Cached features make it possible to regenerate figures and tables without reprocessing the raw data or retraining the models.

4.6 VISUAL REPORTING

The final figures were generated with matplotlib and seaborn using a consistent color palette. All graph labels, tick labels, legends, colorbar labels, and annotations use black text for visibility in the thesis PDF and printed copies. Models use the same colors across all plots: gray for the data-only baseline, blue for the proposed PINN, green for LSTM, and orange-red for CNN. Bearing runs also use fixed colors. In the LaTeX thesis, plot files avoid embedded chart titles where possible because figure captions and labels provide the formal titles in the document.

Chapter 5

Results and Discussion

5.1 FEATURE BEHAVIOR OVER BEARING LIFE

The extracted features do not evolve identically across bearing runs. This is expected because different bearings can have different fault initiation times and degradation rates. RMS and peak-to-peak values capture energy growth. Kurtosis and crest factor are more sensitive to impulsive behavior. E_{kin} summarizes energy near the calculated fault-frequency harmonics.

Figures 5.1 to 5.4 show representative feature evolution over normalized lifetime. The plotted feature values are normalized for visualization so that their trends can be compared across bearings.

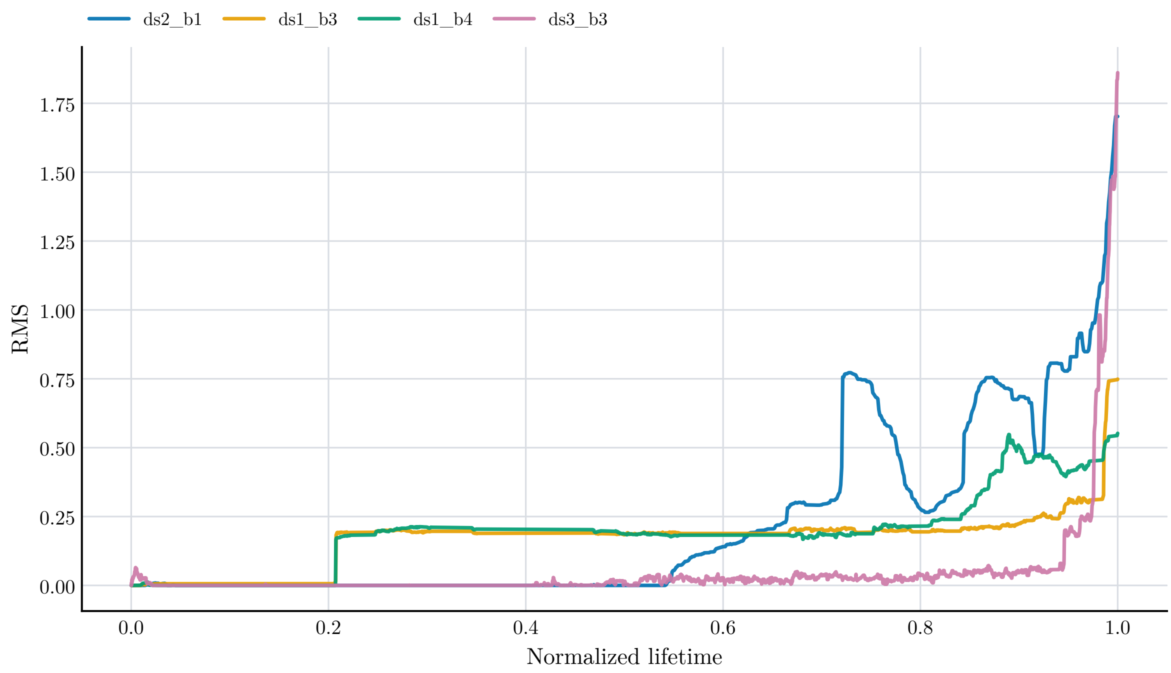

Figure 5.1: RMS evolution over normalized bearing life.

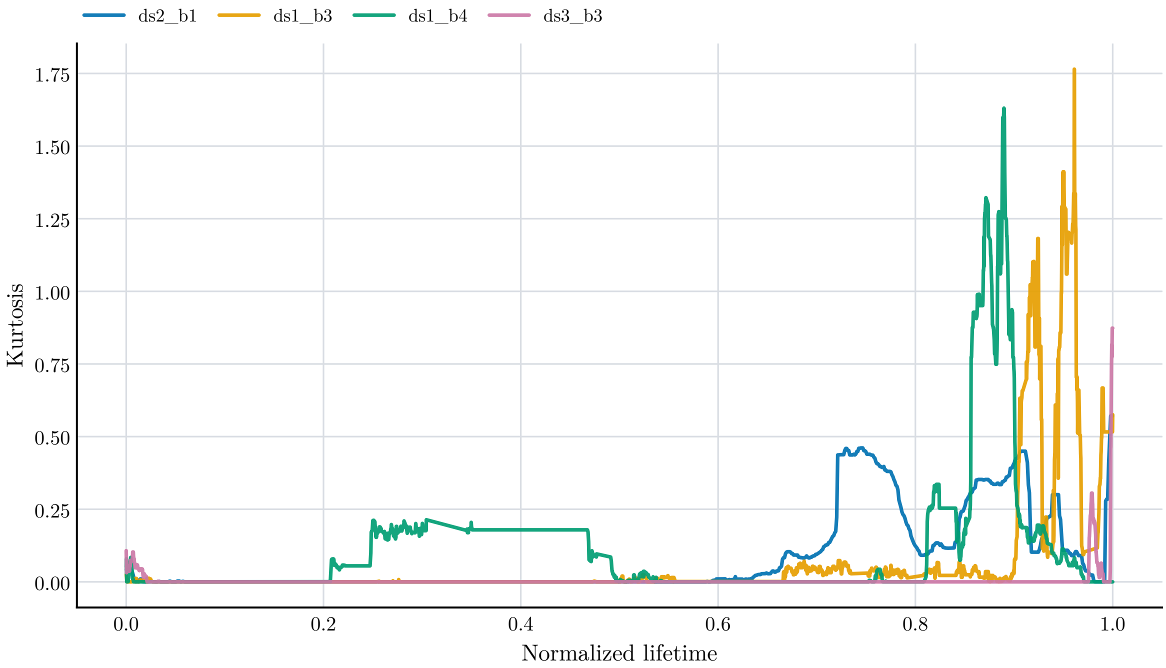

Figure 5.2: Kurtosis evolution over normalized bearing life.

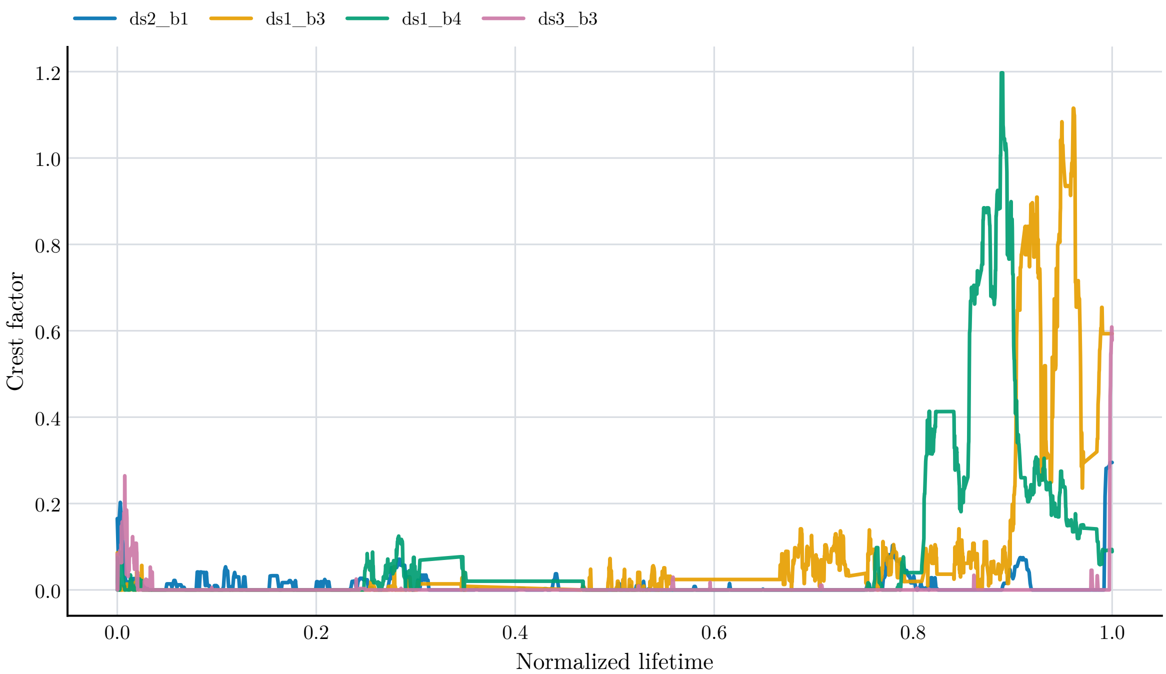

Figure 5.3: Crest factor evolution over normalized bearing life.

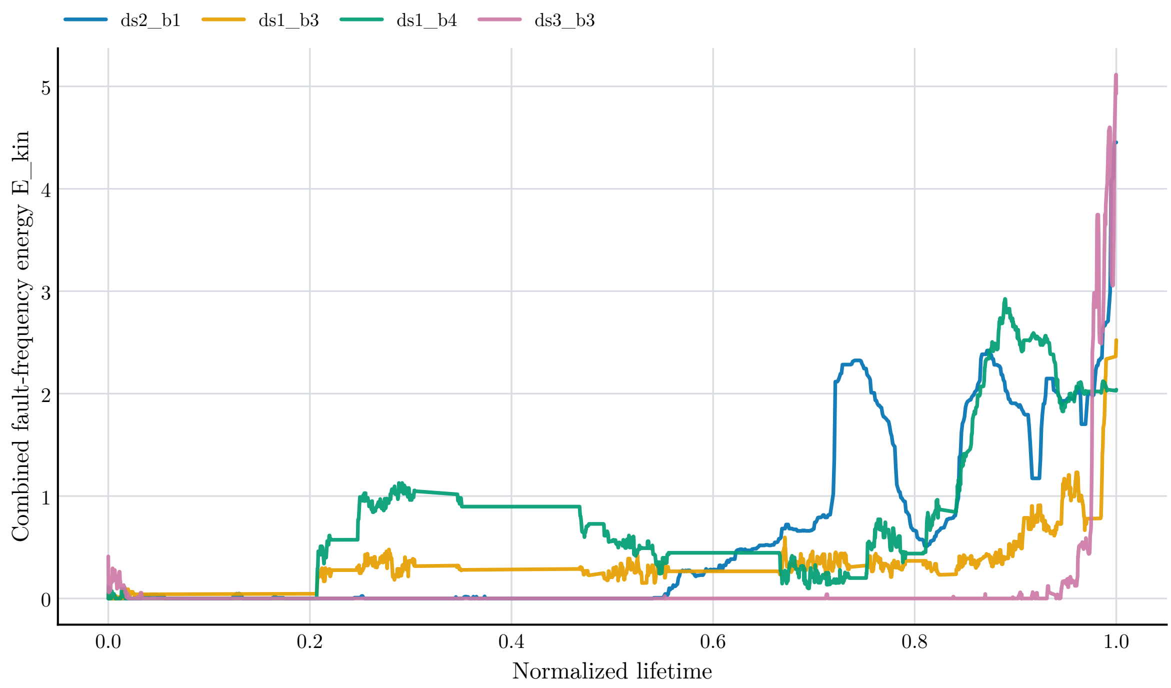

Figure 5.4: Combined fault-frequency envelope energy evolution over normalized bearing life.

The plots show why cross-bearing RUL prediction is difficult. Some features rise smoothly in one run and irregularly in another. A model trained on one degradation shape may misread a different shape. This is the central challenge behind the results that follow.

5.2 MAIN PREDICTION RESULTS

Table 5.1 gives the repeated-seed summary for the final evaluation. The 12 split configurations are trained three times with different random seeds, giving 36 evaluated runs per model. The table reports mean $\pm$ standard deviation across those runs.

Table 5.1: Repeated-seed model performance across 12 split configurations and three seeds per split.

Model	MAE $\pm$ SD	RMSE $\pm$ SD	$R^2 \pm$ SD
Data-only FNN	0.168 ± 0.136	0.209 ± 0.156	0.184 ± 0.885
Proposed PINN	0.178 ± 0.148	0.216 ± 0.170	0.095 ± 0.998
LSTM	0.188 ± 0.072	0.230 ± 0.078	0.256 ± 0.474
CNN	0.203 ± 0.089	0.235 ± 0.095	0.199 ± 0.632

The data-only feed-forward baseline has the lowest mean RMSE, followed closely by the proposed PINN. However, the standard deviations are large compared with the differences between model means. The LSTM baseline has the highest mean R^2, but it does not have the lowest mean RMSE. These results support a conservative interpretation: no model is statistically dominant under the available four bearing trajectories.

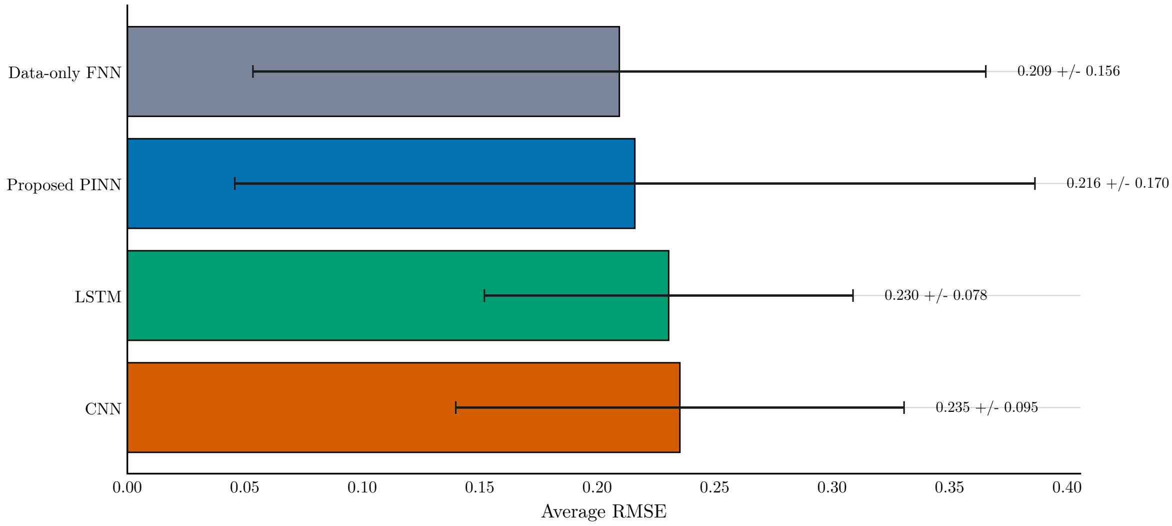

Figure 5.5: Average RMSE by model across 12 split configurations and three random seeds per split.

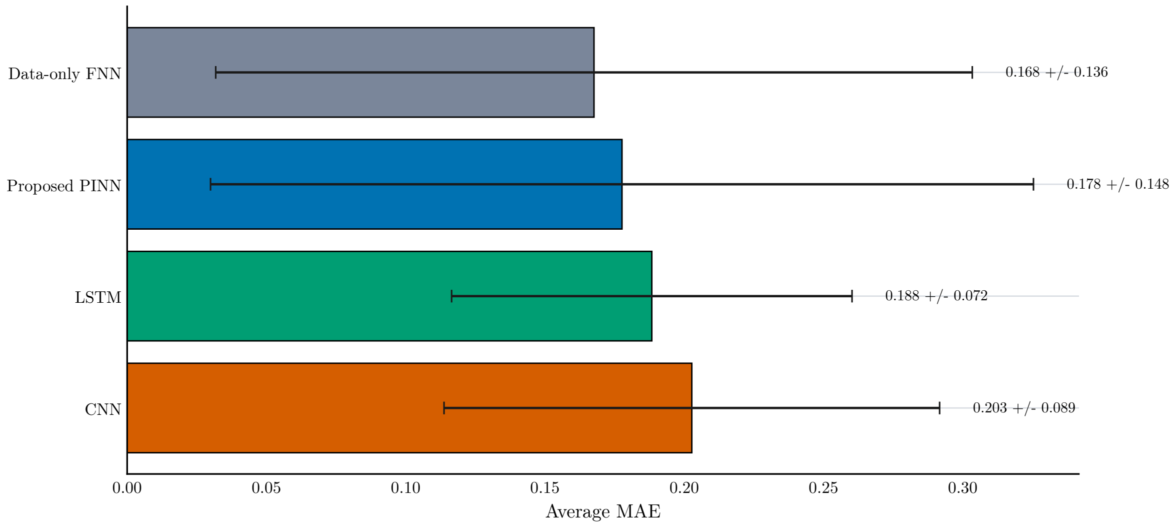

Figure 5.6: Average MAE by model across 12 split configurations and three random seeds per split.

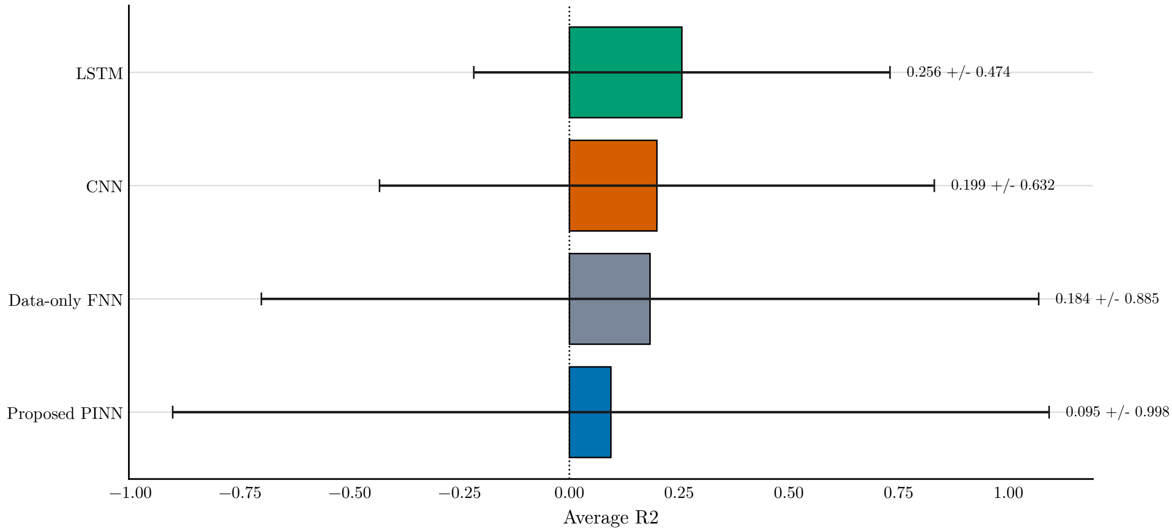

Figure 5.7: Average R^2 by model across 12 split configurations and three random seeds per split.

5.3 STATISTICAL VALIDATION AND UNCERTAINTY

There are still only four full bearing runs in the final data set. This is due to the fact that the snapshots from the same run are not independent of each other and are therefore not formally statistically significant as bearing failures. To overcome this as much as possible in the available data, 12 settings of all the combinations of complete runs of trains, validation and test are enumerated and each train run is repeated with 3 random seeds. This provides 36 rated runs per model, but doesn't provide any more independent bearing failures.

Table 5.2: Statistical validation across 12 split configurations and three repeated seeds.

Model	Splits	Seeds	Runs	Mean RMSE	SD RMSE	Min RMSE	Max RMSE
Data-only FNN	12	3	36	0.209	0.156	0.024	0.468
Proposed PINN	12	3	36	0.216	0.170	0.022	0.484
LSTM	12	3	36	0.230	0.078	0.117	0.355
CNN	12	3	36	0.235	0.095	0.125	0.406

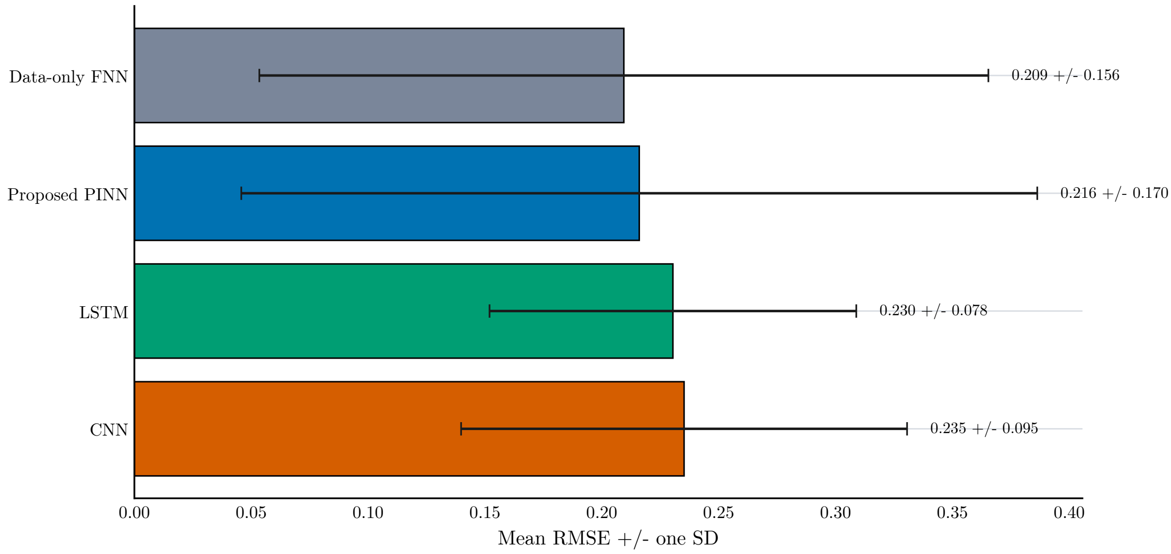

Figure 5.8: Mean RMSE with standard deviation across 12 split configurations and three repeated seeds.

The high standard deviations show that the conclusion depends strongly on both the held-out bearing run and the validation choice. The added validation addresses the earlier lack of repeated-seed reporting, but it also confirms that strong claims of model superiority cannot be made from four bearing trajectories alone.

5.4 EXPERIMENT-WISE COMPARISON

The model ranking by RMSE is given in Table 5.3 and Figure 5.9. The ranking chart uses the mean RMSE over the three repeated seeds within each split configuration. A single model does not win every split.

Table 5.3: Lowest-RMSE model in each split configuration.

Split	Test run	Best model	RMSE ± SD
S01	ds2_b1	LSTM	0.339 ± 0.004
S02	ds2_b1	LSTM	0.339 ± 0.006
S03	ds2_b1	LSTM	0.316 ± 0.027
S04	ds1_b3	Data-only FNN	0.026 ± 0.003
S05	ds1_b3	CNN	0.204 ± 0.011
S06	ds1_b3	Proposed PINN	0.023 ± 0.001
S07	ds1_b4	Data-only FNN	0.035 ± 0.006
S08	ds1_b4	CNN	0.178 ± 0.005
S09	ds1_b4	Proposed PINN	0.045 ± 0.011
S10	ds3_b3	CNN	0.134 ± 0.009
S11	ds3_b3	Data-only FNN	0.146 ± 0.004
S12	ds3_b3	LSTM	0.128 ± 0.017

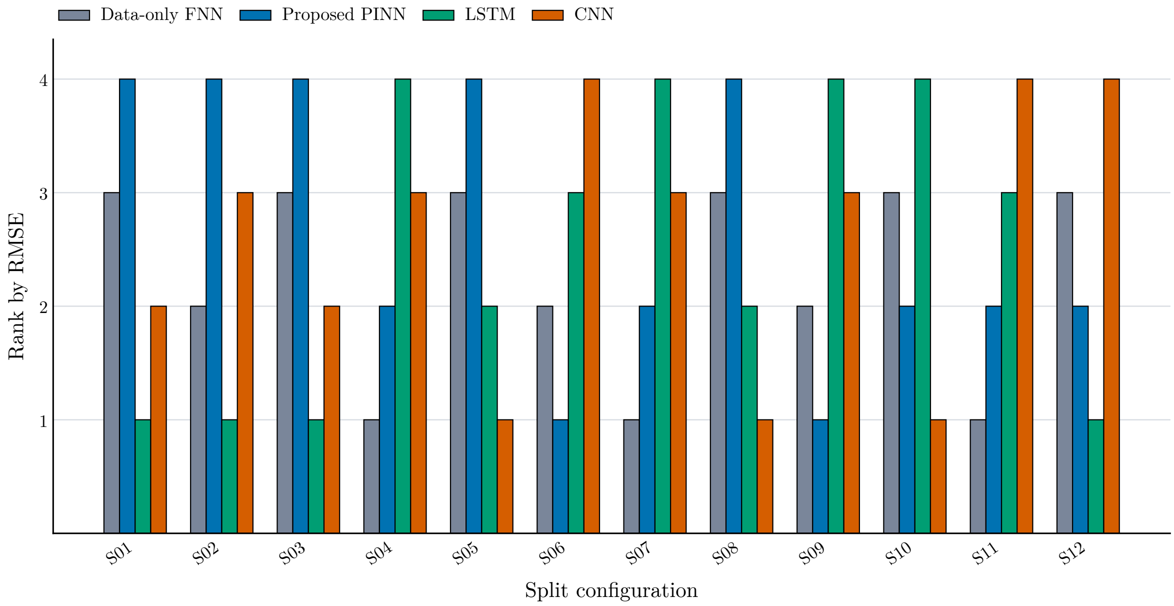

Figure 5.9: Model ranking by mean RMSE for each split configuration. Lower rank is better.

The metric comparison charts in Figures 5.10 to 5.12 show the same split sensitivity from the three evaluation metrics.

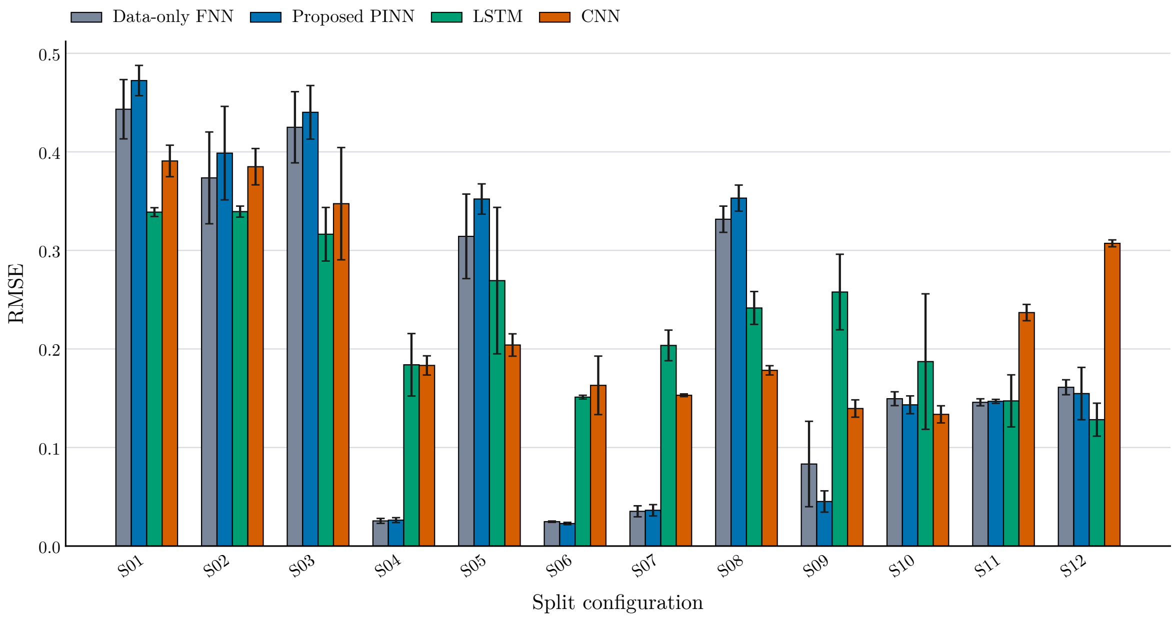

Figure 5.10: RMSE by model and split configuration.

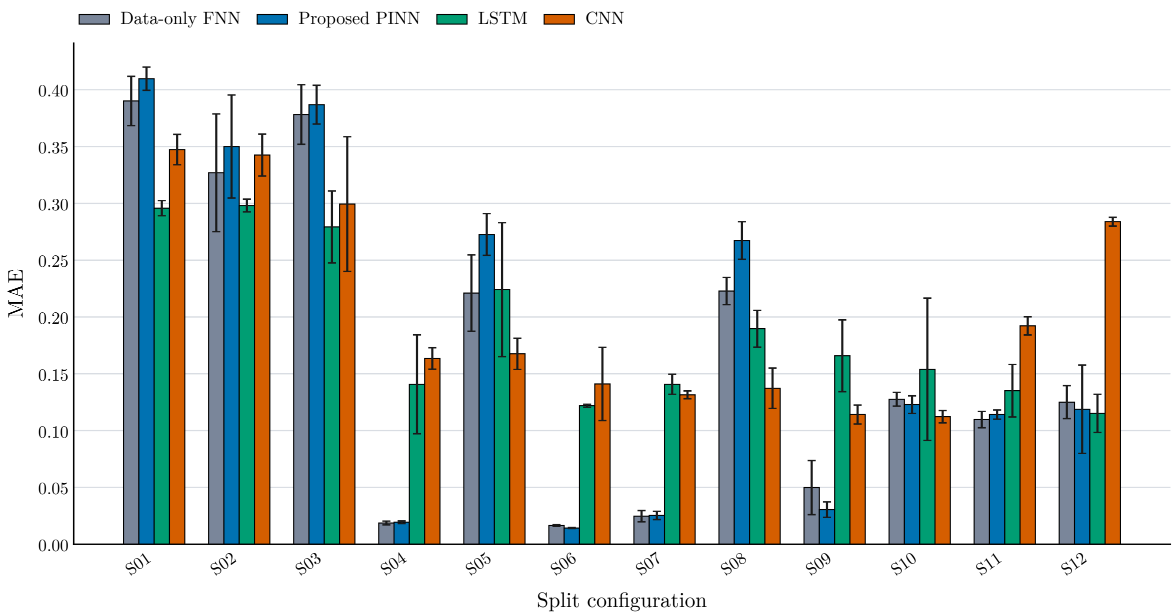

Figure 5.11: MAE by model and split configuration.

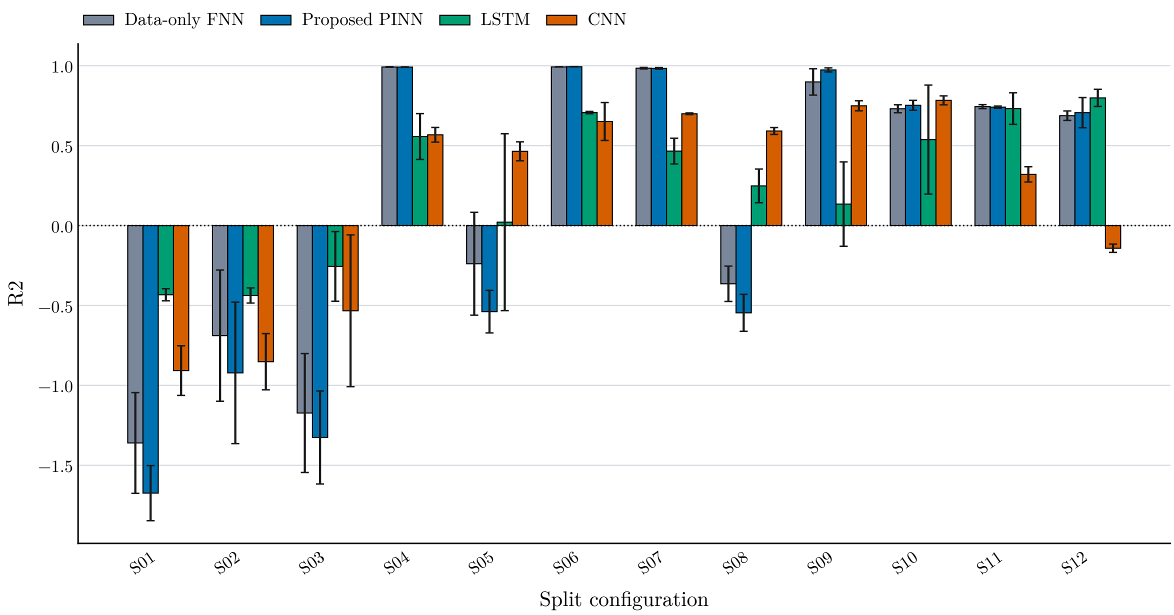

Figure 5.12: R^2 by model and split configuration.

5.5 TRUE VERSUS PREDICTED RUL

Figures 5.13 to 5.16 show representative true-versus-predicted RUL curves from the first seed repeat. Four split configurations are shown so that each selected test bearing appears once.

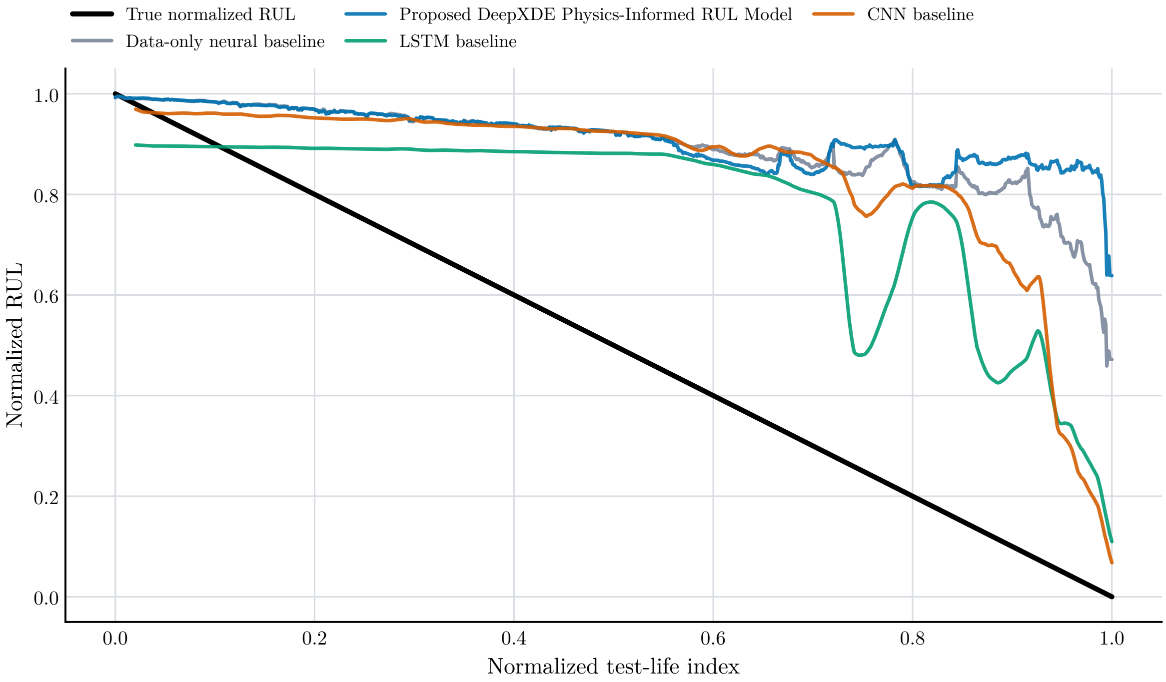

Figure 5.13: True and predicted normalized RUL for split S01 with held-out `ds2_b1`.

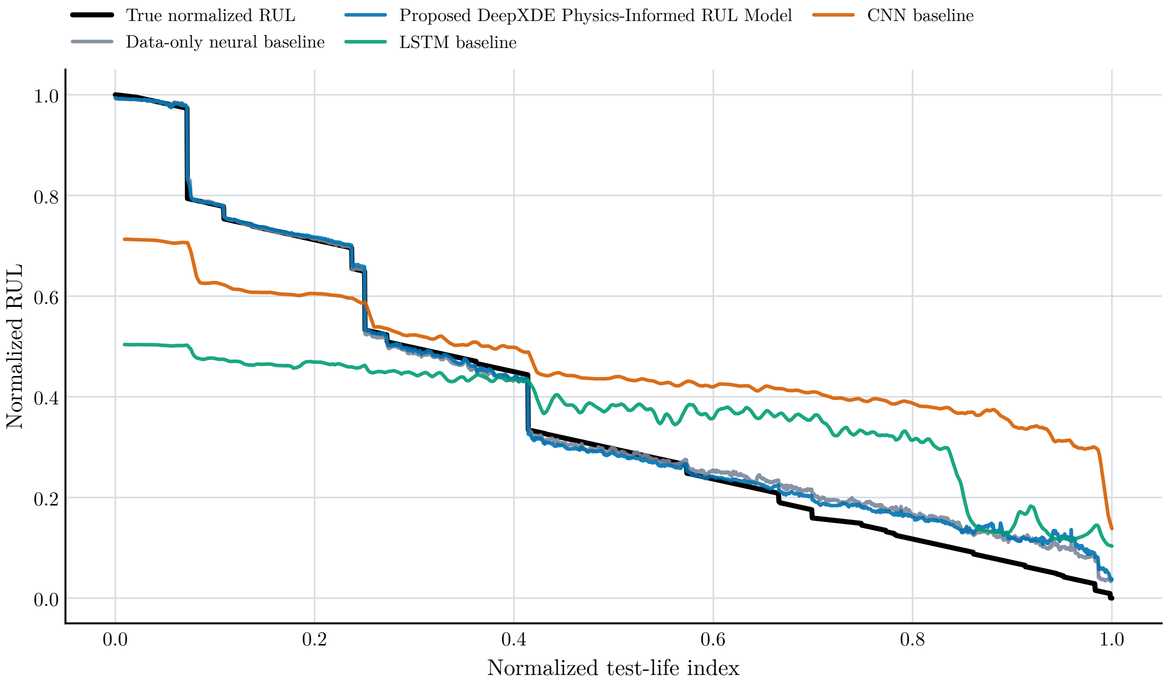

Figure 5.14: True and predicted normalized RUL for split S04 with held-out `ds1_b3`.

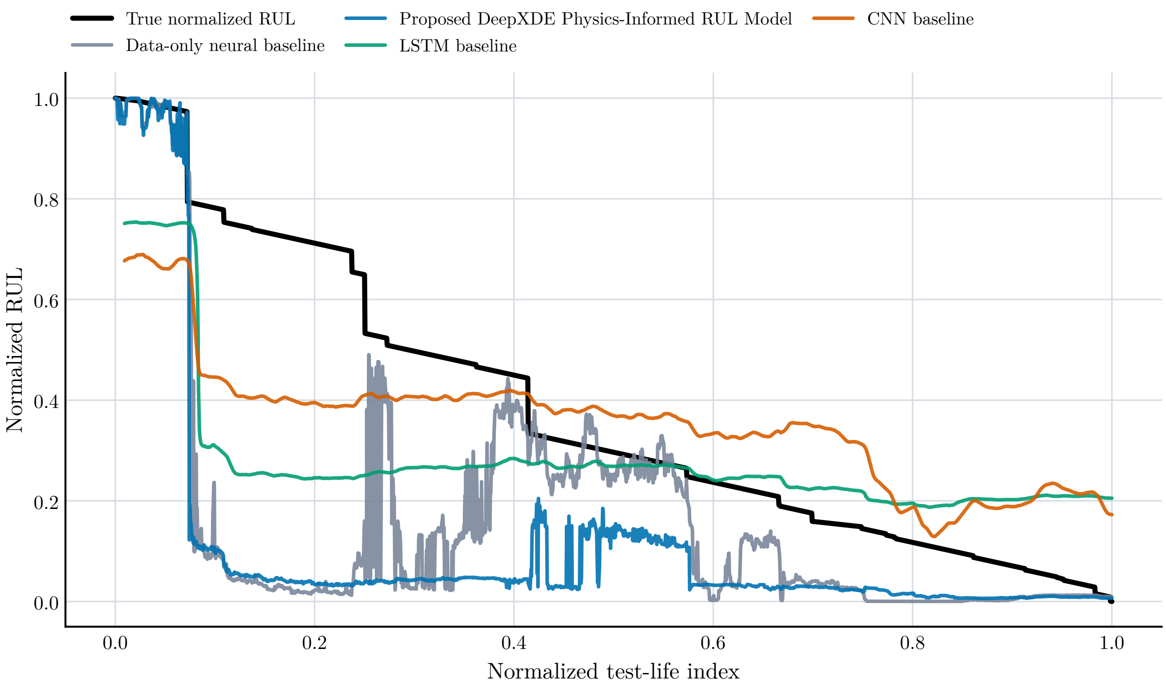

Figure 5.15: True and predicted normalized RUL for split S08 with held-out `ds1_b4`.

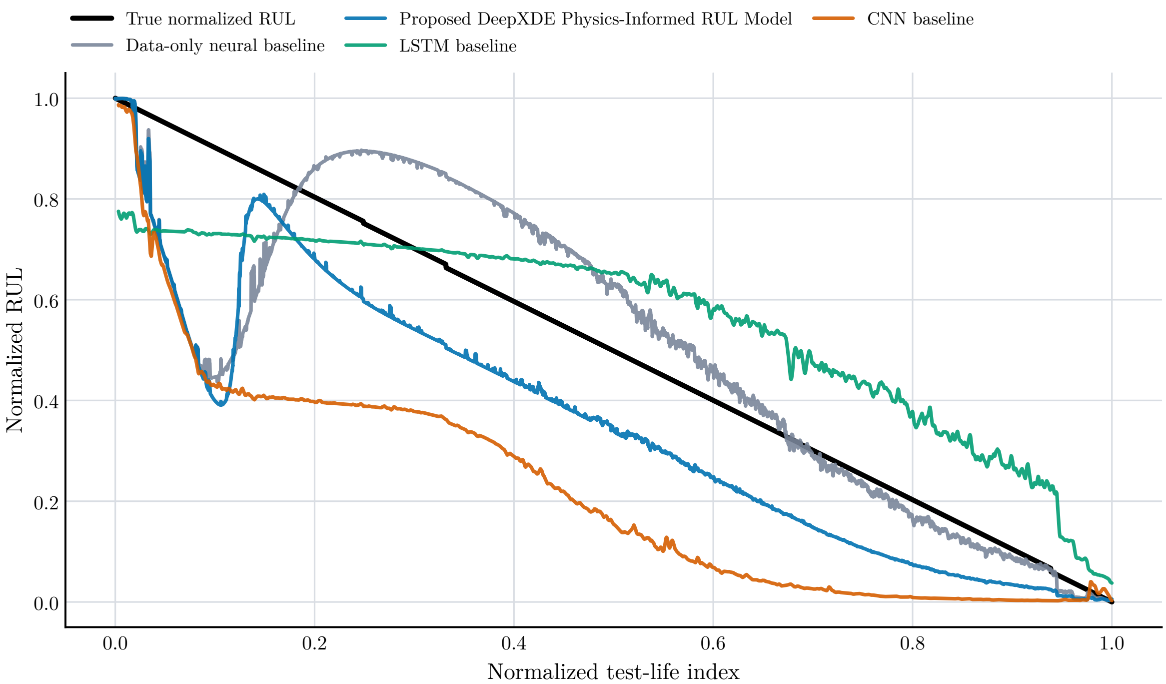

Figure 5.16: True and predicted normalized RUL for split S12 with held-out `ds3_b3`.

Split choice can influence the model's behavior significantly, as indicated by the prediction curves. This is consistent with the repeated-seed tables: uncertainty is generated by the limited number of independent bearing trajectories, and by assignment of the remaining bearing trajectories to validation and training.

5.6 ERROR ANALYSIS

Figure 5.17 shows mean smoothed absolute prediction error over normalized test life across the 12 split configurations. Figure 5.18 shows the signed error distribution, and Figure 5.19 shows 95% block-bootstrap RMSE intervals by split configuration.

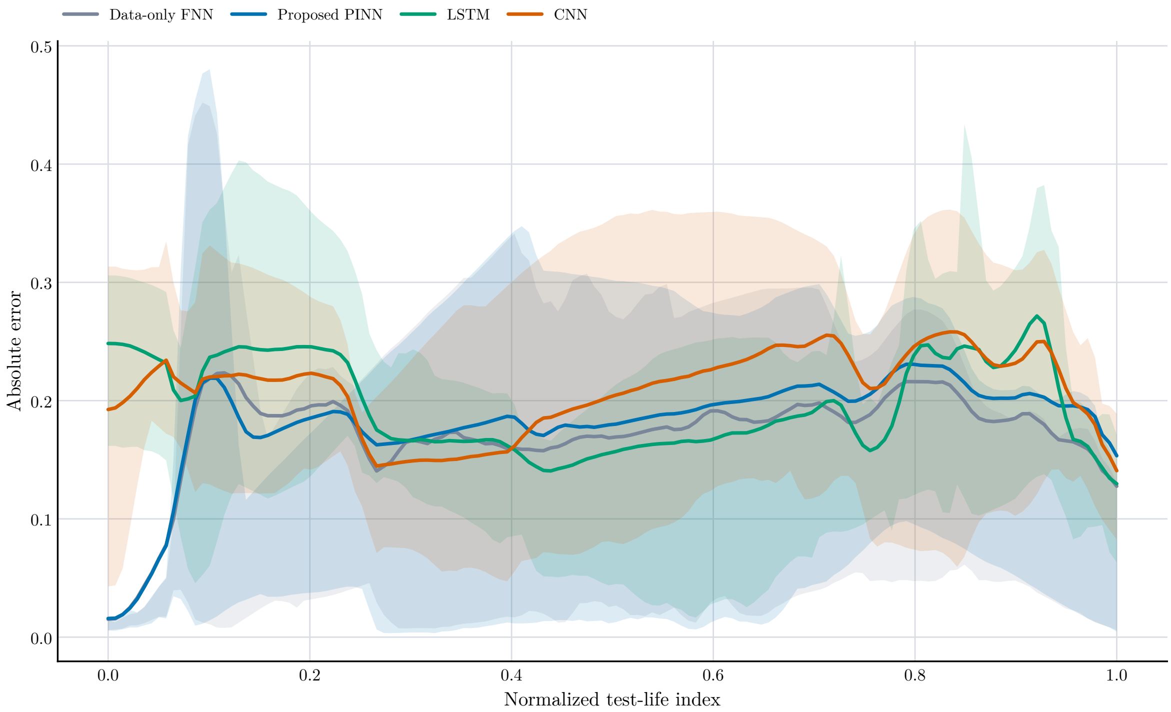

Figure 5.17: Mean smoothed absolute prediction error over normalized test life across the 12 split configurations.

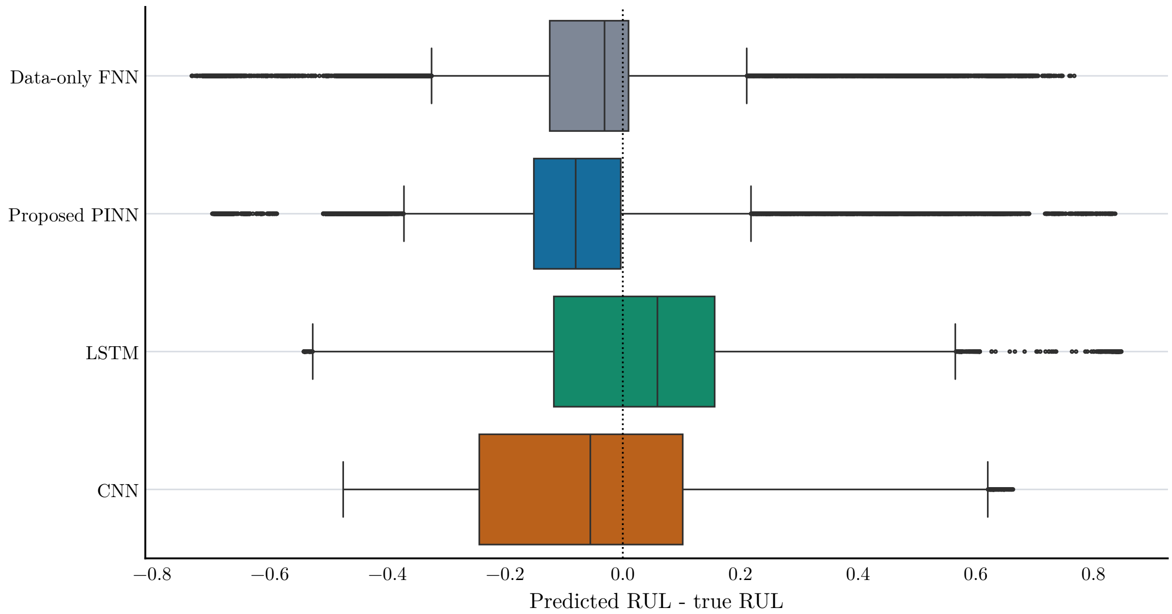

Figure 5.18: Signed prediction error distribution. Positive error means the model overestimated normalized RUL.

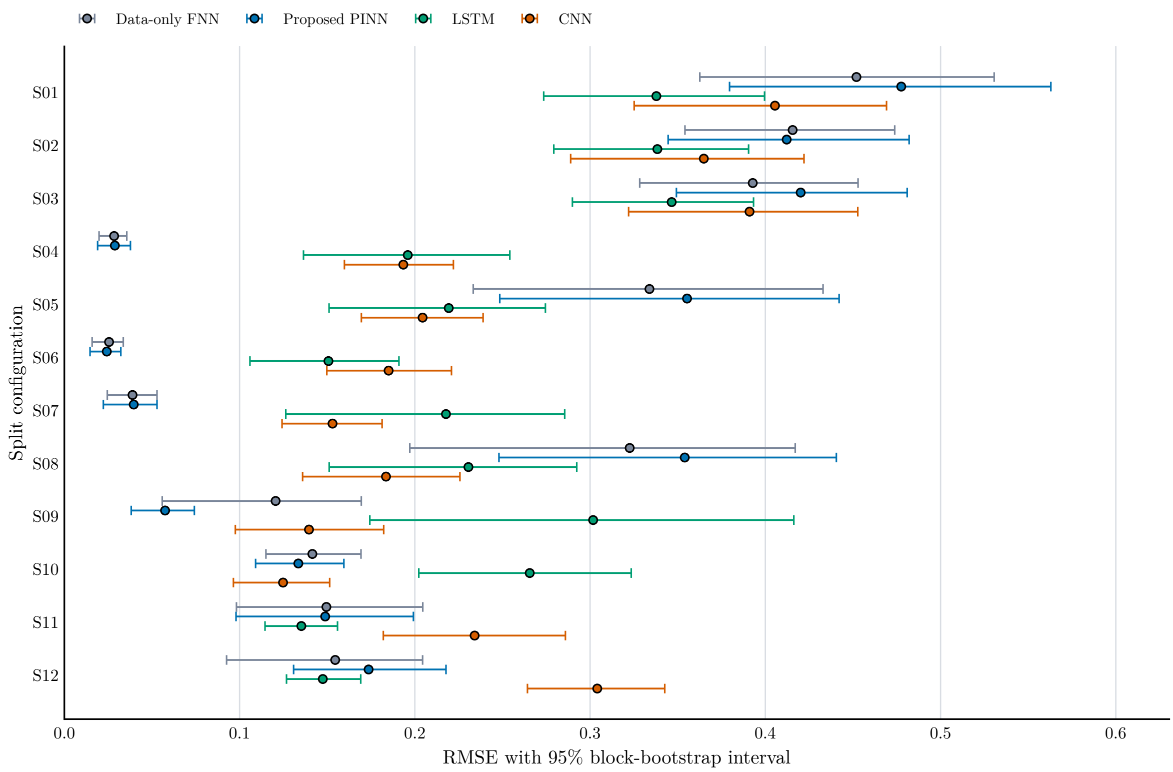

Figure 5.19: RMSE uncertainty intervals by split configuration from block-bootstrap resampling of contiguous prediction-error blocks.

The uncertainty intervals are not to replace further run to failure bearings. They reveal the level of prediction error confidence of test trajectories within the available ones. The large gap between the different intervals in the challenging split settings underscore the lack of statistical significance that can be drawn from 4 bearing runs.

5.7 DISCUSSION OF THE PHYSICS-INFORMED MODEL

There are two types of physical regularisation in the PINN: monotonicity residual and spectral prior with fault-frequency energy. The two assumptions are reasonable engineering, but neither is complete.

The monotonicity residual is based on the premise that RUL should not grow as life goes on. That's right at the life-level, but the model observes noisy features. A strong focus on smooth monotonic behavior may not allow the model to learn local changes in the signal, if the feature pattern of a held out bearing is different than the training bearings. This is minimized by implementing a tolerance and a small weight, but still depends on the split.

The spectral prior makes the assumption that the more severe combined fault-frequency energy the more damage. This has been physically possible but not a measure of actual damages. It can also be influenced by the timing of fault development, the load-zone behavior, the direction of the sensor, and resonance, noise. If E_{kin} increases in one run at the wrong time or not at all, then a model that was learned in a different run may be applied at the wrong time in the first run.

The key idea is that physics-informed learning does not necessarily produce better learning due to the inclusion of engineering information. The previous should be in the vicinity of the actual degradation process to be of assistance. This thesis examines the investigated weak prior in some splits and finds that the prior is not stable enough to be clearly superior in the 12-configuration repeated-seed evaluation.

5.8 PRACTICAL IMPLICATIONS

When it comes to engineering maintenance, the best option is to err on the side of caution. From these results none of the tested models should be used as ultimate RUL predictors. The data-only baseline shows the smallest mean RMSE, and the proposed PINN and sequence baselines are comparable and are dependent on the sequence. For a practical system, there would be more data needed for run-to-failure, more operating conditions, uncertainty estimates and a clear maintenance decision rule.

The work is still helpful in providing a direction. Interpretability is crucial via physics-informed features, and the PINN framework is still promising. Future work should focus on enhancing the physical model, provide more stringent degradation constraints, optimize the physics-loss weights and test on additional bearing runs and datasets.

Chapter 6

Conclusion

6.1 SUMMARY

This thesis aimed at exploring the normalized remaining useful life prediction of some selected runs of rolling-element bearings from the IMS run-to-failure data set. The original notebook-based workflow was refactored into a local Python project that reads raw folders in `data/raw` and extracts the time domain and envelope spectral features; generates normalized labels for the RUL; constructs a PCA based health indicator; trains four neural models; and generates reproducible tables and figures.

The proposed model is a DeepXDE physics-informed neural network with weak residuals, using the RUL physics-informed monotonic behavior and envelope energy of fault frequencies as weak residuals. It was evaluated against a data-only feed-forward neural baseline, and two sequence baselines, an LSTM sequence baseline and a CNN sequence baseline. All the evaluations were done with 12 full-run train-validation-test splits and three repeated random seeds per split.

The lowest mean RMSE was obtained by the data-only feed-forward baseline (mean 0.209 ± 0.156) in 36 runs evaluated for each of the models. The proposed physics-informed model followed at 0.216 ± 0.170, the LSTM baseline at 0.230 ± 0.078, and the CNN baseline at 0.235 ± 0.095. The proposed PINN was therefore competitive in some of the split configurations but the large standard deviations do not allow for statistically convincing superiority.

6.2 CONTRIBUTIONS

This thesis has several important contributions, which are outlined as follows:

1. Local RUL prediction pipeline (for selected IMS bearing run-to-failure data)
2. A documented feature extraction technique that uses time-domain vibration features and bearing fault-frequency envelope energy.
3. A Principal Component Analysis (PCA) health-indicator analysis for consistency assessment of degradation of the extracted features.

4. DeepXDE Physics-Informed RUL Model Based on Monotonicity and Fault-Frequency Energy Priors.

5. A comparison with a feed-forward, LSTM and CNN baseline models for 12 complete-run split configurations.

6. Limited bearing-run sample size is a consequence of repeated-seed statistical validation and block-bootstrap uncertainty analysis.

6.3 LIMITATIONS

The significant constraint is the relatively low number of complete failure runs. The final evaluation extends the design space to 12 split design configurations and three seeds per split, however only four selected bearing trajectory is used. When multiple snapshots are obtained from a single trajectory, these are time dependent and do not substitute for further independent bearing failures. The results are therefore to be regarded as preliminary proof, rather than as statistically solid proof of the superiority of the model.

Other limitations remain. The RUL target is not an independently measured failure threshold but rather a normalization of the target based on the final timestamp. The physical prior is very weak, not even a full contact-fatigue model, lubrication model, temperature effect or a crack-growth law are included. The model only focuses on engineered features and not the raw waveforms. Last but not at all least, the results are obtained from one local implementation and a single set of hyperparameters.

6.4 FUTURE WORK

Ideally, future work should be carried out on more powerful physics-informed formulations. Some potential directions are: uncertainty-aware RUL prediction, Weibull or crack-growth inspired lifetime priors, multi-task learning for prediction of health indicator and RUL, and a learned degradation state that is restricted to monotonic damage growth. The experiments should also be repeated on the PRONOSTIA and other bearing data sets, in order to minimise the risk of relying on a single data set.

The feature pipeline can be enhanced as well. There may be information that is not being captured that can be represented using time-frequency, adaptive envelope bands, and bearing-specific resonance selection and raw waveform sequence models. The comparison would be more statistically sound if more careful tuning of the hyperparameters was done, larger repeated-seed studies were conducted, and more run-to-failure bearings were used.

6.5 FINAL CONCLUSION

The successful experiments demonstrate the potential of the physics-informed bearing RUL prediction yet indicate that it is not a trivial task. The tested weak physics-informed model in this thesis was not clearly better than the data-only baseline in mean RMSE, but showed variability in the results and was not clearly better across the split configurations. Physical interpretation has to be combined with statistically sound learning models of bearing condition assessment, and a practical bearing prognostics system needs to be evaluated by testing with more independent run-to-failure data, before it can be claimed as a strong solution.

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Appendices

APPENDIX A

Cost estimation of the project:

Items	**Cost (Taka)**
Software:	
i) Python 3.11 (programming environment)	Free/Open Source
ii) PyTorch and DeepXDE (modeling and PINN libraries)	Free/Open Source
iii) NumPy, pandas, SciPy, scikit-learn, matplotlib, seaborn	Free/Open Source
iv) `uv` (dependency management)	Free/Open Source
v) LaTeX/MiKTeX (typesetting)	Free/Open Source
Databases and Data Sources:	
i) IMS Bearing Dataset (NASA Prognostics Data Repository)	Free/Publicly available
ii) Reference papers and literature (secondary data)	Free/Publicly available
Hardware/Computing Resources:	
i) Laptop (AMD Ryzen 7 5800H, NVIDIA RTX 3060 Laptop GPU) used for model training and analysis	Already Available
ii) Local storage for datasets, features, and figures	Already Available
Miscellaneous:	
i) Documentation and report writing tools (LaTeX)	Free/Open Source
ii) Version control (Git/GitHub)	Free
iii) Reference management (Zotero/Mendeley)	Free
Total Estimated Cost:	**Negligible**

APPENDIX B

Gantt chart of the project:

APPENDIX C

The originality and AI reports generated by Turnitin are attached below.

MAHDEE HASSAN

ORIGINALITY REPORT

6%	3%	4%	3%
SIMILARITY INDEX	INTERNET SOURCES	PUBLICATIONS	STUDENT PAPERS

PRIMARY SOURCES

1	Submitted to Chittagong University of Engineering and Technology Student Paper	<1 %
2	Mohammed Chalouli, Nasr-eddine Berrached, Mouloud Denai. "Intelligent Health Monitoring of Machine Bearings Based on Feature Extraction", Journal of Failure Analysis and Prevention, 2017 Publication	<1 %
3	arxiv.org Internet Source	<1 %
4	Jacek Dybała. "Diagnosing of rolling-element bearings using amplitude level-based decomposition of machine vibration signal", Measurement, 2018 Publication	<1 %
5	coek.info Internet Source	<1 %
6	Submitted to Liverpool John Moores University Student Paper	<1 %
7	ojs.istp-press.com Internet Source	<1 %

0% detected as AI

The percentage indicates the combined amount of likely AI-generated text as well as likely AI-generated text that was also likely AI-paraphrased.

Caution: Review required.

It is essential to understand the limitations of AI detection before making decisions about a student's work. We encourage you to learn more about Turnitin's AI detection capabilities before using the tool.

0 AI-generated only 0%
Likely AI-generated text from a large-language model.

0 AI-generated text that was AI-paraphrased 0%
Likely AI-generated text that was likely revised using an AI-paraphrase tool or word spinner.

APPENDIX D

CO-PO-K-P-A Mapping

(This part will be filled/checked by the supervisor)

PO1	**Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO2	**Problem analysis:** Identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO3	**Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
PO4	**Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO5	**Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
PO6	**The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
PO7	**Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
PO8	**Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9	**Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
PO10	**Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO11	**Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO12	**Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Title of the Project: Physics-Informed Neural Networks for Remaining-Useful-Life Prediction of Rolling-Element Bearings

Student ID: 2003101

		CO-PO-K-P-A Mapping																															
		Program Outcomes												**Knowledge Profile**								**Complex Engineering Problem Solving**							**Complex Engineering Activities**				
		PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**	**K1**	**K2**	**K3**	**K4**	**K5**	**K6**	**K7**	**K8**	**P1**	**P2**	**P3**	**P4**	**P5**	**P6**	**P7**	**A1**	**A2**	**A3**	**A4**	**A5**
		Needs complex engineering problem solution (P1 + another P)																											Related to project				
Course No.	Course Title	Engineering Knowledge	Problem Analysis	Design/Development of Solutions	Investigation	Modern Tool Usage	The Engineers & Society	Environment and Sustainability	Ethics	Individual and team work	Communication	Project Management and Finance	Life-long Learning	Science	Math	Engineering fundamentals	Engineering specialization	Design	Technology	Society	Research	Knowledge K3,K6,K8	Wide ranging/conflicting	No obvious solution	Infrequent issues	Outside problems	Diverse groups	Many components	Range of resources	Level of interaction	Innovation	Consequences	Familiarity
ME 498	Project and Thesis																																